

ZHERDEVKA, Russian FederationWMO: **340470**Lat: **51.833N**Lon: **41.483E**Elev: **483**StdP: **14.44**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.3	-8.1	-20.5	1.8	-13.9	-13.9	2.6	-7.7	26.1	26.2	22.4	21.2	5.7	300	0.544

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	89.0	67.2	85.6	66.0	82.6	64.8	70.2	83.4	68.6	80.6	67.2	78.3	7.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	97.8	75.0	64.5	92.6	73.3	63.1	87.9	72.0	34.4	83.8	33.1	81.2	31.9	78.3	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	20.8	17.8	15.7	DB	-16.8	94.4	5.2	3.9	-20.6	97.2	-23.6	99.4	-26.5	101.6	-30.3	104.4
(5)				WB	-18.0	72.7	6.2	1.9	-22.5	74.1	-26.1	75.3	-29.6	76.3	-34.2	77.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.9	17.3	18.1	28.0	46.2	59.5	65.8	68.8	67.1	57.1	44.2	30.9	21.8	
	DBStd	21.22	12.25	12.51	9.14	8.90	9.05	7.33	5.55	6.60	7.92	8.87	10.45	11.73	
	HDD50	4486	1013	894	682	174	23	1	0	0	25	224	574	876	
	HDD65	8143	1478	1314	1147	564	217	81	23	54	255	646	1023	1341	
	CDD50	2250	0	0	0	61	318	476	584	529	238	43	1	0	
	CDD65	431	0	0	0	0	47	106	142	118	18	0	0	0	
	CDH74	4263	0	0	0	12	489	1007	1324	1219	210	2	0	0	
	CDH80	1285	0	0	0	0	117	300	429	407	32	0	0	0	
Wind	WSAvg	7.4	8.3	8.6	8.7	8.3	6.9	6.3	5.8	6.0	6.3	7.1	8.0	8.7	
Precipitation	PrecAvg	19.8	1.4	1.2	1.1	1.3	1.8	2.6	2.2	1.5	2.3	1.7	1.5	1.5	
	PrecMax	26.0	2.8	2.2	2.2	5.2	4.0	5.9	6.2	5.0	5.6	3.5	3.5	2.9	
	PrecMin	13.5	0.4	0.3	0.1	0.2	0.4	0.8	0.1	0.1	0.1	0.5	0.1	0.5	
	PrecStd	3.5	0.6	0.6	0.6	1.1	1.0	1.4	1.3	1.1	1.6	0.9	1.1	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.6	38.9	54.0	76.0	87.0	90.3	94.0	93.1	83.4	71.1	54.9	45.8	
		MCWB	34.7	37.1	45.2	55.1	65.3	66.2	69.1	68.4	61.2	56.5	52.0	44.4	
	2%	DB	34.8	35.9	47.2	70.9	83.4	86.5	89.2	88.6	79.8	65.7	50.6	39.1	
		MCWB	33.5	34.4	41.9	54.5	62.8	67.0	68.6	66.8	60.4	54.4	48.4	37.7	
	5%	DB	33.7	34.7	42.9	66.5	79.8	83.5	84.6	84.3	75.8	61.0	47.1	36.0	
		MCWB	32.7	33.3	38.7	52.4	62.1	66.1	66.6	64.7	59.5	53.3	45.1	34.6	
	10%	DB	32.5	33.5	39.2	62.0	75.6	80.1	81.2	80.8	71.2	56.8	43.6	34.2	
		MCWB	31.7	32.3	36.2	49.9	60.5	65.2	65.4	64.3	57.9	51.1	41.7	33.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.5	37.4	46.9	59.1	68.0	71.8	72.7	71.9	66.0	60.3	52.8	44.3
			MCDB	36.0	38.6	52.2	68.1	81.0	83.4	86.5	85.0	77.5	67.1	54.8	45.9
2%		WB	33.8	34.9	42.8	55.9	65.6	69.7	70.0	69.5	63.1	57.2	48.7	37.9	
		MCDB	34.4	35.5	46.9	65.9	79.1	81.8	83.4	84.0	72.2	62.8	50.0	39.0	
5%		WB	32.8	33.5	39.2	53.6	63.6	68.0	68.5	67.5	61.1	54.1	45.3	34.9	
		MCDB	33.6	34.3	42.5	64.7	76.0	78.4	80.6	79.3	71.2	59.6	46.8	35.7	
10%		WB	31.7	32.4	36.5	51.3	61.6	66.2	67.1	65.9	59.1	51.5	42.0	33.3	
		MCDB	32.5	33.5	38.8	61.3	73.0	77.3	77.8	77.0	69.7	56.2	43.5	34.1	
Mean Daily Temperature Range	5% DB	MDBR	10.0	11.6	12.3	18.0	19.7	18.6	19.3	20.9	18.8	13.5	9.2	9.2	
		MCDBR	6.3	7.8	15.3	25.1	24.7	23.6	24.5	26.1	25.6	21.3	12.3	7.2	
		MCWBR	6.2	7.4	10.9	12.9	10.0	9.4	9.1	9.3	11.0	12.4	10.7	6.9	
	5% WB	MCDBR	6.4	7.8	14.5	22.4	21.7	19.9	20.4	21.9	20.8	18.2	11.2	7.4	
		MCWBR	6.4	7.7	10.6	12.8	10.3	9.6	8.9	9.9	10.9	11.9	10.3	7.2	
Clear-Sky Solar Irradiance	taub	0.304	0.341	0.366	0.434	0.398	0.390	0.417	0.438	0.386	0.362	0.338	0.308		
	taud	2.108	2.007	2.070	2.105	2.314	2.355	2.278	2.209	2.340	2.348	2.260	2.148		
	Ebn at Noon	211	234	257	253	272	274	264	250	251	232	200	187		
	Edh at Noon	24	36	41	45	38	37	40	40	32	26	21	20		
All-Sky Solar Radiation	RadAvg	293	612	1082	1372	1840	1911	1884	1609	1078	557	248	165		
	RadStd	66	98	165	111	160	178	149	135	132	84	53	45		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (7 neighbors)	+1.26	N/A	N/A	N/A	N/A	N/A	N/A	-247	+220	N/A